INCH-POUND
MIL-PRF-1/1658D
22 October 2019
SUPERSEDING
MIL-PRF-1/1658C
16 June 2003

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON
TYPE 8789

Inactive For New Design
after 19 June 2015

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and MIL-PRF-1.

DESCRIPTION: Hydrogen.
See figure 1.
Mounting position: Vertical, base down.
Weight: 9 pounds nominal.

ABSOLUTE RATINGS:

Parameter Unit	Ef V ac	epy kv	ep x kv	Ebb kV dc	egy v	egx v	Ecc V dc	ib a	Ib A dc	Ip Aac	tj μs
Maximum	6.6	33 1/	33 2/	---	3/	650	---	2,000	4.0	72	0.01
Minimum	6.0	7	5% epy	3.5	---	---	---	---	---	---	---
Test condition	6.3	33	---	---	1,100	---	0	---	---	---	---

Parameter Unit	prr ---	dik/dt a/μs	Pb ---	tk sec	TA °C	Cooling ---	Eres Vac
Maximum	---	10,000	30×10^9	5/	+75	6/	5.5 8/
Minimum	---	---	---	900	-55	---	3.5
Test condition	465	---	---	900	---	7/	9/

See footnotes at end of table I.

GENERAL:

Qualification - Required.

This specification sheet uses accept on zero defect sampling in accordance with MIL-PRF-1, table III.

Service-life guarantee – 500 hours minimum tube operating time.

Marking – Tubes sold under service-life guarantee shall be marked with the contract number and with the number of tube operating hours guaranteed.

AMSC N/A

FSC 5960

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TABLE I. Testing and inspection.

Inspection	Method MIL-STD-1311	Conditions	Symbol	Limits Min	Limits Max	Unit
Qualification inspection						
Operation (3) 7/ 10/ 11/ 12/	----	TA = 75°C, except t = 5 hours	egy	----	1,100	v
Life-test (2) provisions 7/ 11/ 13/ 14/	----	epy = 25 kv; prr = 740; Ib = 4.0 Adc (min); egy = 1,200 v; Zg = 50 Ω (min); t = 1,000 hours	----	----	----	----
Life-test (2) end points:						
Operation (1)	----	Zg = 50 Ω (min)	egy	----	1,200	v
DC anode voltage (1)	3246	egy = 1,200 v; Zg = 50 Ω (min)	Ebb	----	3,000	V dc
Time jitter	3261	egy = 1,200 v; Zg = 50Ω (min)	tj	----	0.01	μs
Conformance inspection, part 1						
Dimensions	----	See figure 1	----	----	----	----
Heater current (cathode)	3241		If	22	35	Aac
Heater current (reservoir)	3241	Eres = 4.5 V ac	Ires	3.0	10.0	Aac
DC anode voltage (1) 15/	3246	Eres = 105 percent of optimum value	Ebb	----	2,500	V dc
Instantaneous start 7/ 10/ 16/	----	Eres = 105 percent of optimum reservoir voltage; epy = 24 kv (min)	----	----	----	----
Operation (1) 7/ 10/ 11/ 17/ 18/	----	epy = 36 kv; t = 1 hour	egy	----	1,100	v
Reservoir voltage (1) 19/	----	Operation (1)	Eres	3.5	5.5	V dc
Emission 20/	3251	ik = 2,000 a; tp = 5.0 μs ± 10 percent; prr = 60 pps ± 10 percent; tr = 0.5 μs (max)	egk	----	300	v

See footnotes at end of table.

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TABLE I. Testing and inspection Continued.

Inspection	Method MIL-STD-1311	Conditions	Symbol	Limits		Unit
				Min	Max	
Conformance inspection, part 2						
Dimensions	----	See figure 1	----	----	----	----
DC anode voltage (2) 21/	3246	Eres = 95 percent of optimum value	Ebb	----	2,500	V dc
Operation (1A) 7/ 11/ 18/ 22/	----	Operation (1)	egy	----	1,100	v
Operation (2) 7/ 11/ 13/ 17/ 18/	----	epy = 25 kv; t = 1 hour; prr = 740; Ib = 4.0 A (min)	egy	----	1,100	v
Operation (2A) 7/ 11/ 18/ 22/	----	Operation (2)	egy	----	1,100	v
Anode delay time	3256	Operation (1A) t = 2 minutes	tad	----	1.0	µs
Anode delay time drift 23/	3256	Anode delay time	Δtad	----	0.1	µs
Time jitter 24/	3261	Operation (1A); except epy = 7.0 kv	tj	----	0.01	µs
Sweep frequency vibration	----	4/	----	----	----	----
Shock	----	4/	----	----	----	----
Conformance inspection, part 3						
Life-test (1) provisions 7/ 10/ 11/ 14/ 25/	----	Group C; egy = 1,200 v; Zg = 50 Ω (min); t = 500 hours	----	----	----	----
Life test (1) end points:	----					
Operation (1)	----	Zg = 50 Ω (min)	egy	----	1,200	v
DC anode voltage (1)	3246	egy = 1,200v; Zg = 50 Ω (min)	Ebb	----	3,000	V dc
Time jitter	3261	egy = 1,200v; Zg = 50Ω (min)	tj	----	0.01	µs

See footnotes at end of table.

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TABLE I. Testing and inspection - Continued.

1/ Instantaneous starting is not recommended. When it is absolutely necessary, the maximum permissible epy is 22 kv and it shall not be attained in less than 0.04 second.

2/ In pulsed operation, the peak inverse voltage shall not exceed 5.0 kv during the first 25 μs following the pulse.

3/ The driver pulse, measured at the tube socket with thyratron grid disconnected: 1,300 volts minimum; 2,500 volts maximum; tr =0.35 μs maximum; grid pulse duration 2.0 μs minimum. Impedance of drive circuit 10 to 25 ohms.

4/ Shock and vibration tests.

(a) Shock tests. Shock tests shall be performed in accordance with MIL-STD-202-213, except as noted. Shock machines of the same type as specified but with rubber impact pads for more reproducible shock acceleration patterns may be used. The tube shall be mounted on suitable rigid fixtures in the position described below by means of its mounting flange. The tube shall not be clamped, braced, or supported by means other than its mounting flanges.

The tube shall be subjected to three blows in each of the following positions:

Position of tube elevator assembly	Intensity and duration of blow
Vertical - base down	50 G, 11 ms
Horizontal - center line through heater terminals, vertical	50 G, 11 ms
Horizontal - center line through heater terminals, horizontal	50 G, 11 ms

The tests shall be performed with both heater and reservoir at substantially full operating temperatures. Means for detecting heater current vibrations and shorts between anode and grid and between grid and cathode shall be provided. No temporary or permanent change in heater or reservoir currents or temporary or permanent shorts between electrodes shall occur.

(b) Vibration tests. Vibration tests shall be performed as specified in MIL-STD-202-201. Tube shall be vibrated as described for 2 hours each in the vertical direction, in the horizontal direction parallel to the heater connections, and in the horizontal direction perpendicular to the heater connections.

(c) After shock and vibration tests, the tube shall meet all the conformance inspection, part 1 and part 2 test requirements. An aging period not to exceed 1 hour is permitted after mechanical tests.

(d) The anode terminal extension may be removed during performance of the shock and vibration tests.

(e) For the purpose of defining a production “lot”, quantities of types 7390 and 8789 produced concurrently, shall be combined to constitute a “lot”. Tests may be performed prior to installation of permanent leads and adaptive mounting plate.

5/ Standby operation, with heater and reservoir voltage on, is not recommended. Where standby operation is necessary, provisions shall be made to operate the tube at full equipment conditions for a minimum of 2 hours during each 12-hour period of standby.

6/ Cooling is preferred at high duty conditions.

7/ Approximately 15 cfm of air may be directed at grid seal and lower envelope.

8/ The optimum reservoir voltage for operation in accordance with operation (1) condition is inscribed on the base of the tube and must be held within ± 2.5 percent. Applications involving other operation conditions will necessitate the re-determination of the optimum reservoir voltage.

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TABLE I. Testing and inspection - Continued.

9/ Except as otherwise specified, adjust reservoir voltage to value indicated on the tube within ± 5 percent.

10/ The anode circuit constants shall be chosen so that at epy = 33.0 kv under resonant charging conditions; ib = 2,000 amps minimum; dik/dt shall be 10,000 amps/μs minimum; tp = 2.5 μs ± 1 0 percent; prr = 500 pps minimum. The tube shall be tested in the test circuit shown on figure 4.

Warning: These conditions are specified only for the purpose of determining circuit constants. The actual operating voltage and repetition rates for each test are specified in the conventional manner under the particular conditions or under the general test conditions. Tests performed at repetition rates less than the resonant charging rate shall be made with a hold-off diode in the charging circuit.

Grid pulse measured at tube socket with thyratron grid disconnect: tr = 0.35 μs minimum; tp = 2.0 μs maximum; internal impedance of driver: 75 ohms minimum.

11/ There shall be no evidence of deleterious anode heating during the test.

12/ The tube shall be capable of operating under the following conditions:

(a) Apply cathode heater voltage and adjust reservoir voltage to 95 percent of the optimum reservoir voltage and allow both to warm up for 15 minutes.
(b) Raise anode voltage directly to the specified voltage at a rate of not less than 30 kv per minute.
(c) The tube shall operate at the specified anode voltage for a total of 2 hours which may include one interruption. A total of four interruptions at any anode voltage is permitted provided that the total test time required to perform this part of the operation test does not exceed 135 minutes.
(d) After the above operation at full anode voltage, increase the reservoir voltage to 105 percent of the optimum value. The tube shall continue to operate for an additional 2 hours with not more than one interruptions.
(e) After the above operation at full anode voltage, decrease the reservoir voltage to optimum value. The tube shall continue to operate for an additional hour without interruption.

13/ The anode circuit constants shall be chosen so that at epy = 25.0 kv under resonant charging conditions; ib = 1,000 minimum; rate of rise of dik/dt shall be 5,000 amps/μs minimum; tp = 5.0 μs ± 10 percent; prr = 500 minimum. The tube shall be tested in the test circuit shown on figure 4.

Warning: These conditions are specified only for the purpose of determining circuit constants. The actual operating voltage and repetition rates for each test are specified in the conventional manner under the particular conditions or under the general test conditions. Tests performed at repetition rates less than the resonant charging rate shall be made with a hold-off diode in the charging circuit.

Grid pulse measured at tube socket with thyratron grid disconnected: tr = 0.35 μs minimum; tp = 2.0 μs maximum; internal impedance of driver: 75 ohms minimum.

14/ During every 96-hour period, the life test shall be shut off for a minimum of 30 minutes. With qualifying activity approval occasional shutdowns for testing or plant closing to cover off hours is permissible under life test.

15/ This test shall be performed in the same circuit as operation (1) test and within 60 seconds following the conclusion of the operation (1) test. The anode voltage shall be increased until the tube starts to conduct current. Ebb shall be below the specified limit.

16/ This test shall be the first test performed after the holding period. The tube shall operate satisfactorily on pushbutton starting within three attempts when the epy is applied to the tube under test in such a manner as to rise from 0 to 24 kv minimum within 0.04 second. (The filter in the rectifier shall be designed so that the epy reaches at least 9.0 kv within 0.015 second.) The intervals between successive attempts to instantaneously start the tube shall be not less than 10 or more than 30 seconds. Any tube failing to start within three attempts will be considered a failure.

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TABLE I. Testing and inspection - Continued.

17/ The tube shall be capable of operating under the following conditions:

(a) Apply cathode heater voltage and adjust reservoir voltage to 105 percent of the optimum value and allow both cathode and reservoir to warm up for 15 minutes.
(b) Raise anode voltage directly to the specified voltage at the rate of not less than 30 kv per minute.
(c) The tube shall operate at the specified anode voltage for a total of 60 minutes which may include one interruption. A total of six interruptions at any anode voltage is permitted provided that the total test time required to perform the operation test does not exceed 120 minutes.

18/ No anode or grid voltage other than that required for an instantaneous start test shall have been applied to the tube for a minimum period of 2 hours prior to this test.

19/ The optimum reservoir voltage shall be determined for each tube under these conditions, shall be inscribed permanently on the base of each tube, and shall be within the limits specified for reservoir voltage (1).

20/ With the anode floating, the positive pulse shall be applied to the grid of the tube. Measure the voltage between grid and cathode not more than 2.5 μs maximum after the beginning of the current pulse. The average voltage shall not rise during the last 4.0 μs. As an alternate, the test may be conducted by connecting the grid to the anode through a 1-ohm resistance and reading epk. The limit for this reading will be the same as that for egk in the emission test.

21/ This test shall be performed in the same circuit as operation (1A) test and within 60 seconds following the conclusion of the operation (1A) test. The anode voltage shall be increased until the tube starts to conduct current. Ebb shall be below the specified limit.

22/ The tube shall be capable of operating at the following conditions:

(a) Apply cathode heater voltage and adjust reservoir voltage to 95 percent of the optimum reservoir voltage and allow both to warm up for 15 minutes.
(b) Raise anode voltage directly to the specified voltage at a rate of not less than 30 kv per minute.
(c) The tube shall operate at the specified anode voltage for a total of 60 minutes which may include one interruption. A total of six interruptions is permitted provided that the total test time required to perform the operation test does not exceed 120 minutes.

23/ During the interval between 2 and 60 minutes of the anode delay time test, the Δtad relative to the tad value observed on the anode delay time test shall not exceed the specified value.

24/ The tj should be measured at 50 percent of the pulse amplitude and shall be not greater than the amount specified after 60 seconds of operation.

25/ With qualifying activity approval the manufacturer may provide, in accordance with MIL-PRF-1, service-life guarantee, in lieu of performing life testing. Life test endpoints specified shall apply to service-life guarantee conformance as well as life test conformance. The number of hours of system-deployed, accumulated tube-operating time shall be approved by the qualifying activity and shall be a minimum of 500 hours. Service-life guarantee shall define tube operating time and not time from purchase or delivery. Tubes sold under service-life guarantee shall be marked with contract number and with the number of tube operating hours guaranteed. The qualifying activity may restore life testing requirements when service performance or tube reliability indicate it is justified.

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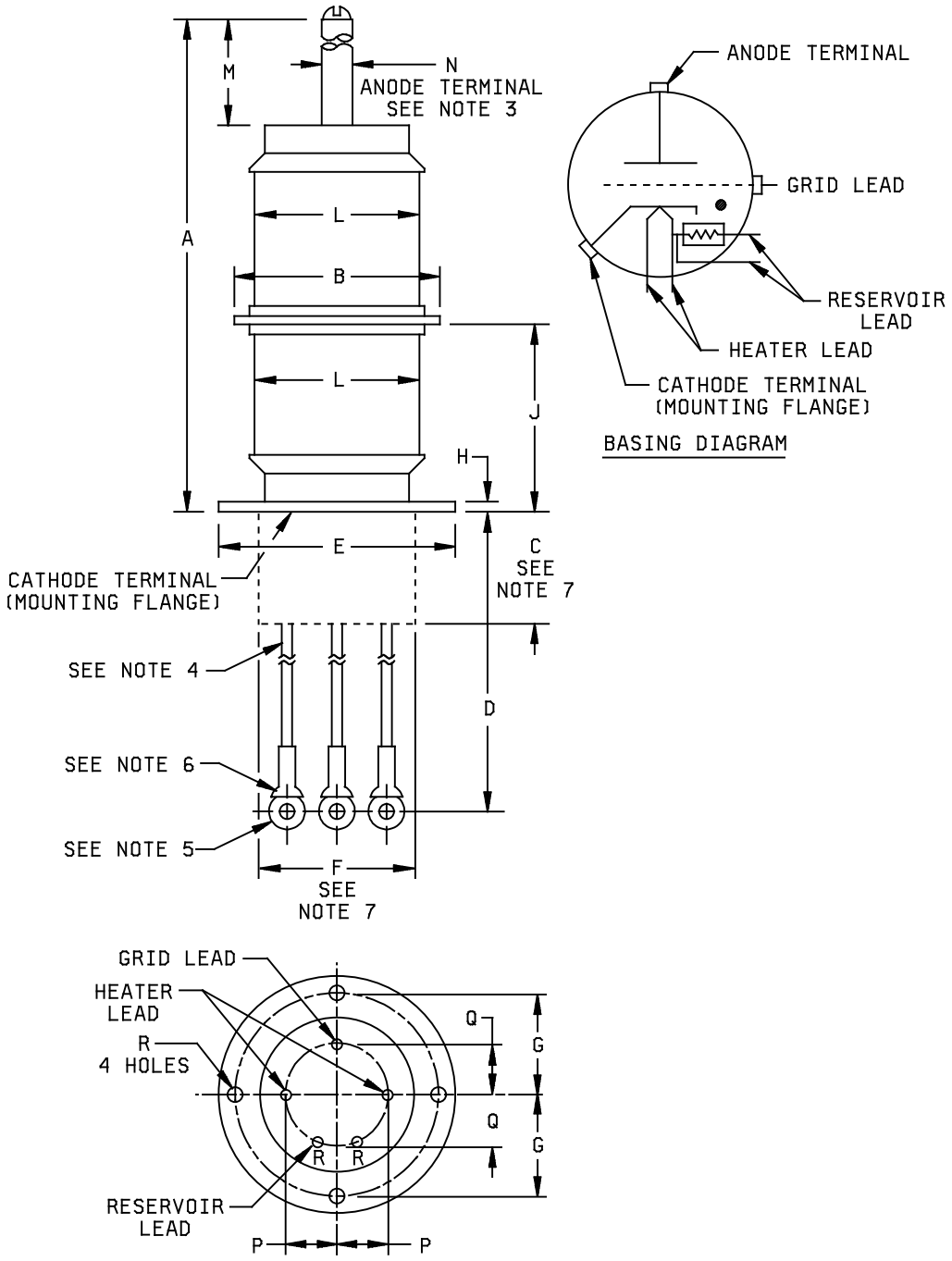


FIGURE 1. Outline drawing of electron tube type 8789.

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Ltr	Dimensions (see note b)			
	Millimeters		Inches	
Conformance inspection, part 1				
	Min	Max	Min	Max
A	428.63	479.43	16.875	18.875
B	----	142.88	----	5.625
Conformance inspection, part 2				
D	304.80	330.20	12.000	13.000
E	214.12	217.42	8.430	8.560
G	96.60	97.10	3.803	3.823
Conformance inspection, part 3 (see note i) (see notes c, e, f, g, and h)				
C	----	76.20	----	3.000
F	----	160.32	----	6.312
J	95.25	146.05	3.750	5.750
L	111.13	117.48	4.375	4.625
M	----	285.75	----	11.250
N	15.24	16.51	0.600	0.650
P	25.40	50.80	1.000	2.000
Q	22.23	47.63	0.875	1.875
R	7.85	8.00	0.309	0.315
Nominal dimensions (see note a)				
H	3.18		0.125	

NOTES:
1. Dimensions without tolerances are for information only and are not required for inspection purposes.
2. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Anode terminal shall be No. 10-32 UNC-2A roundhead machine screw 3/8 inch (9.53 mm) long with washer. The anode terminal extension may be removed for shipment.
4. Leads shall be flexible. Lead connections shall be suitably identified. Lead connections, lead identification, lead insulation, and lead insulation sleeving shall be inspected under periodic check and shall be as follows:

Lead	Lead insulation		
	Color	Type	Sleeving Color
H (heater)	Yellow	HA-1	Yellow
G (grid)	Yellow or Green	HA-1	Green
R (reservoir)	Red	HA-1	Red

5. Lead terminal shall be in accordance with figure 2.
6. Insulation sleeving shall be as specified in MIL-I-3190.
7. Dimensions C and F define extent of rigid portion of tube below mounting flange. Clearance for this part of tube shall be provided in equipment.
8. Holes R shall be inspected for orientation by using the base gage outlined on figure 3.
9. These dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. An accept on zero defect sampling plan shall be used, with a sample of three tubes with an acceptance number of zero.

FIGURE 1. Outline drawing of electron tube type 8789 - Continued.

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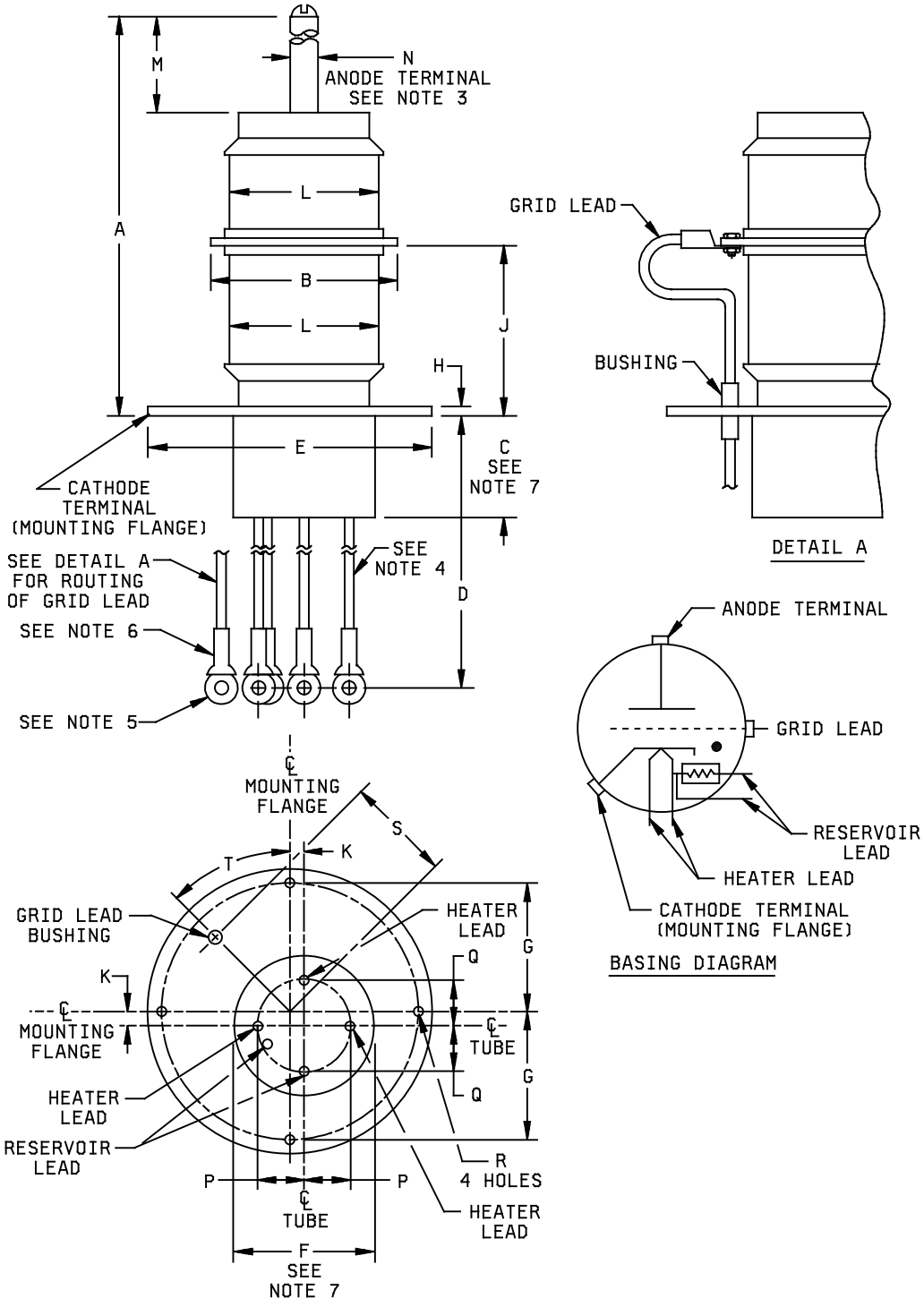


FIGURE 1A. Alternate outline drawing of electron tube type 8789.

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Ltr	Dimensions			
	Millimeters		Inches	
Conformance inspection, part 1				
	Min	Max	Min	Max
A	428.63	479.43	16.875	18.875
B	----	142.88	----	5.625
Conformance inspection, part 2				
D	241.30	266.70	9.500	10.500
E	214.12	217.42	8.430	8.560
G	96.60	97.10	3.803	3.823
Conformance inspection, part 3 (see notes c, e, f, g, h, and i)				
C	----	76.20	----	3.000
F	----	160.32	----	6.312
J	95.25	146.05	3.750	5.750
K	4.75	11.10	0.187	0.437
L	111.13	117.48	4.375	4.625
M	----	285.75	----	11.250
N	15.24	16.51	0.600	0.650
P	25.40	50.80	1.000	2.000
Q	22.23	47.63	0.875	1.875
R	7.85	8.00	0.309	0.315
S	----	80.16	----	3.156
Nominal dimensions (see note a)				
H	3.18		0.125	
T	45° ± 10°			

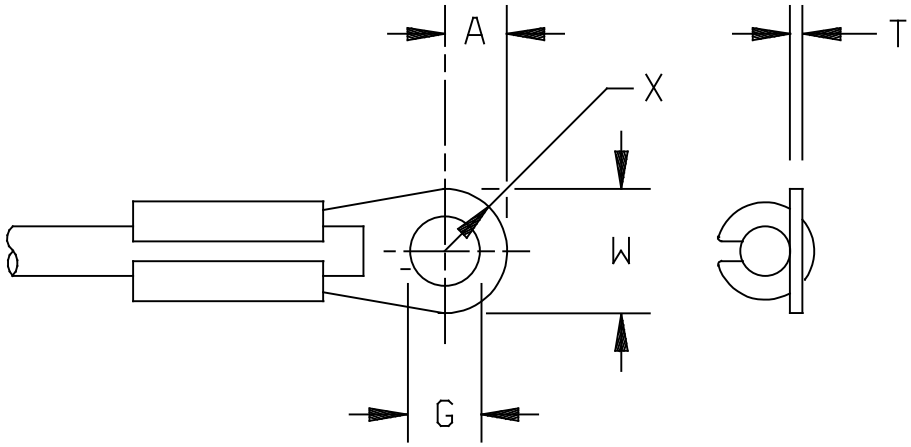
NOTES:
1. Dimensions without tolerances are for information only and are not required for inspection purposes.
2. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Anode terminal shall be No. 10-32 UNC-2A roundhead machine screw 3/8 inch (9.53 mm) long with washer. The anode terminal extension may be removed for shipment.
4. Leads shall be flexible. Lead connections shall be suitably identified. Lead connections, lead identification, lead insulation, and lead insulation sleeving shall be inspected under periodic check and shall be as follows:

	Lead insulation		
Lead	Color	Type	Sleeving Color
H (heater)	Yellow	HA-1	Yellow
G (grid)	Yellow or Green	HA-1	Green
R (reservoir)	Red	HA-1	Red

5. Lead terminal shall be in accordance with figure 2.
6. Insulation sleeving shall be as specified in MIL-I-3190.
7. Dimensions C and F define extent of rigid portion of tube below mounting flange. Clearance for this part of tube shall be provided in equipment.
8. Holes R shall be inspected for orientation by using the base gage outlined on figure 3.
9. These dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. An accept on zero defect sampling plan shall be used, with a sample of three tubes with an acceptance number of zero.

FIGURE 1A. Alternate outline drawing of electron tube type 8789 - Continued.

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Lead	Lug designation	G				W	
		Inches		Millimeters		Inches	Millimeters
		Min	Max	Min	Max	Max	Max
Reservoir, Grid	#10	0.187	0.207	4.75	5.26	0.395	10.03
Heater	0.250	0.260	0.313	6.60	7.95	0.605	15.37

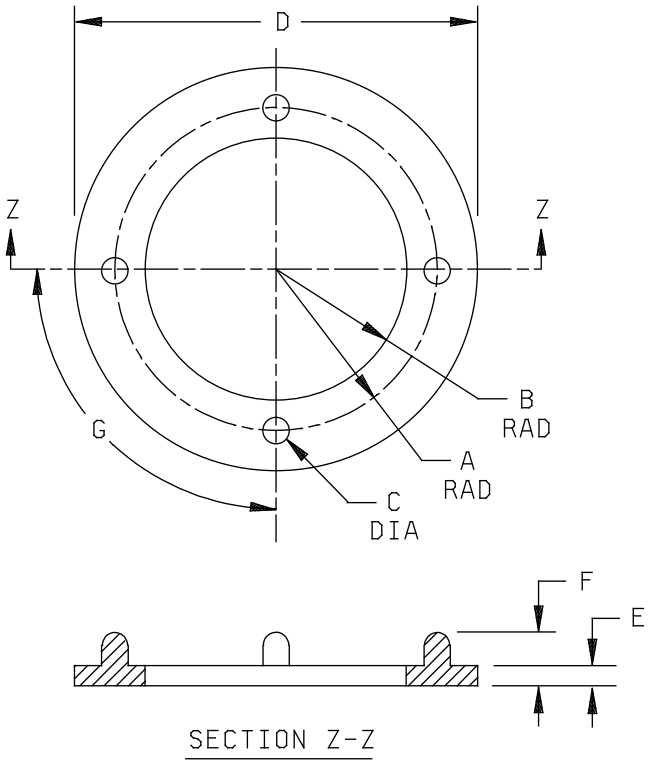
Lead	Lug designation	A		X		T	
		Inches	Millimeters	Inches	Millimeters	Inches	Millimeters
		Max	Max	Min	Min	Max	Max
Reservoir, Grid	#10	0.200	5.08	0.275	6.99	0.060	1.52
Heater	0.250	0.305	7.75	0.380	9.65	0.060	1.52

NOTES:

1. Dimensions are in inches.
2. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. There shall be no obstruction within the distance X from the center of the lug screw hole.
4. Lugs or terminals shall be permanently marked as follows: H (heater)
   G (grid)
   R (reservoir)

FIGURE 2. Pressure type terminal lugs with insulating sleeves.

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	3.812	3.814	96.82	96.88
B	3.187	3.189	80.95	81.00
C	.2765	.2775	7.02	7.05
D	---	9.00 DIA Nom	---	228.6 DIA Nom
E	.296	.328	7.52	8.33
F	---	.250	---	6.35
G	88°	92°	88°	92°

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 3. Base gage for electron tube type 8789.

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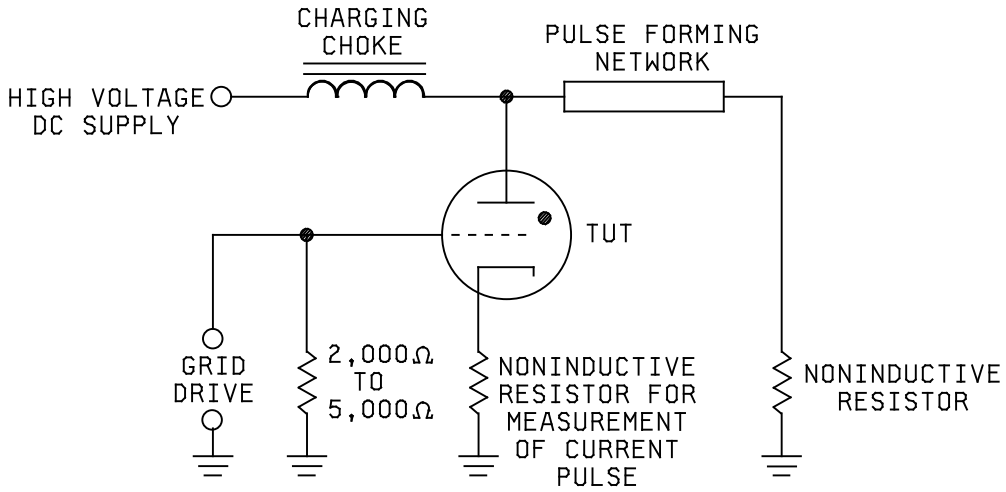


FIGURE 4. Test circuit.

Referenced documents. In addition to MIL-PRF-1, this document references the following:
MIL-STD-202-201
MIL-STD-202-213
MIL-STD-1311
MIL-I-3190

Revision notations. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Air Force - 85
DLA - CC

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(Project 5960-2019-019)

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